GMAT Focus Edition

Complete Pre-Exam Review Handbook

Every key concept, rule & strategy — mapped to the *GMAT Official Guide 2025–2026*

Quantitative Reasoning	21	45 min	Problem-Solving	No calculator
Data Insights	20	45 min	Data Sufficiency, Multi-Source Reasoning, Table Analysis, Graphics Interpretation, Two-Part Analysis	On-screen calculator
Verbal Reasoning	23	45 min	Reading Comprehension, Critical Reasoning	—
Total	64	135 min	Plus one optional 10-minute break	Choose section order

Scoring & format

Total score runs from **205 to 805** in 10-point steps; each of the three sections is scored from **60 to 90**. The exam is **section-adaptive**: questions get harder or easier based on your running performance, so you **cannot skip** a question. Within each section you may bookmark questions and, in the Question Review & Edit step, change up to **three** answers before time runs out. You choose the order of the three sections.

How to use this handbook

This document collects the concepts taught in the Official Guide's review chapters (Ch. 3 Math, Ch. 5 Data Insights, Ch. 7 Verbal) plus the question-type strategies from Ch. 4, 6, and 8 — everything you need to refresh the night before. Chapter numbers are shown beside each heading so you can jump to the source for worked examples. Read it straight through for a full review, or jump to a section. The orange boxes are strategy and trap reminders.

Last-day pacing reminders

- Quant: ~2 min 9 sec per question. Data Insights: ~2 min 15 sec. Verbal: under 2 min.
- If stuck, eliminate what you can, pick the best remaining choice, and move on — never leave a question blank.
- Every question contains all the info you need; do not rely on outside knowledge of the topic.
- Take the 10-minute break. Start with whichever section you're strongest in to build momentum.

1 Quantitative Reasoning

Official Guide Ch. 3 (Math Review) & Ch. 4 — arithmetic and algebra only; no geometry, no calculator

3.1 Value, Order & Factors

Numbers, the number line & absolute value

Number line	Every real number is a point; left of 0 is negative, right is positive. Each number is less than any number to its right. Every real number except 0 is positive or negative.
Between	“x is between 1 and 4” means $1 < x < 4$; “inclusive” means $1 \le x \le 4$.
Absolute value	$\lvert x\rvert$ = distance from 0, always ≥ 0. $\lvert x\rvert = x$ if $x \ge 0$, and $-x$ if $x < 0$. So $\lvert -3\rvert = \lvert 3\rvert = 3$. $\lvert x+y\rvert \le \lvert x\rvert+\lvert y\rvert$.

Factors, multiples & remainders

Integers	{… –2, –1, 0, 1, 2, …}. Consecutive integers differ by 1.
Factor / multiple	If $x = yn$ for integers, y is a factor/divisor of x and x is a multiple of y. 4 and 7 are factors of 28 since $28 = 7\times4$.
Quotient & remainder	Dividing x by y gives unique q, r with $x = yq + r$ and $0 \le r < y$. $28 \div 8 \rightarrow q=3, r=4$. Remainder is 0 iff x is divisible by y. A smaller ÷ larger gives quotient 0 and remainder = the smaller number.
Remainder shortcut	Remainder of $(a+b)\div n$ = remainder of $(ra+rb)\div n$; same for products. Useful for huge numbers.
Even / odd	Even = divisible by 2. Even×anything = even; odd×odd = odd. Even±even and odd±odd = even; even±odd = odd.

Primes, composites & prime factorization

Prime	A positive integer with exactly two divisors, 1 and itself. First primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29… 2 is the only even prime; 1 is NOT prime.
Composite	Integer > 1 that isn't prime; equals a unique product of primes. E.g. $484 = 2^2\cdot11^2$.
Divisibility via primes	x is divisible by y iff x's prime factorization contains every prime in y's, each to at least as high a power. # of factors = product of (each exponent + 1).

Exponents & roots

Definition	x^n = x multiplied n times; x is the base, n the exponent. Squaring a number > 1 enlarges it; squaring a number between 0 and 1 shrinks it.
Square root	$\sqrt{x} \ge 0$ is the nonnegative root; $\sqrt{(x^2)} = \lvert x\rvert$. The square root of a negative number is not real (only real numbers appear on the GMAT).
Cube root	Every real number has exactly one real cube root: $\sqrt[3]{8} = 2$, $\sqrt[3]{(-8)} = -2$.
Power rules	$x^a\cdot x^b = x^{a+b}$; $x^a/x^b = x^{a-b}$; $(x^a)^b = x^{ab}$; $x^0 = 1$; $x^{-a} = 1/x^a$; $(xy)^a = x^a y^a$.

Decimals, place value & scientific notation

Repeating decimals	A finite digit-block repeats forever, shown with a bar. Can be converted to a fraction algebraically.
Scientific notation	One nonzero digit left of the point × a power of 10. Positive exponent → move point right; negative → left. $0.0231 = 2.31\times10^{-2}$.
Operations	Add/subtract: line up decimal points. Multiply: ignore points, multiply as integers, then place point so decimal places = sum of the two inputs'. Divide: shift both points right until divisor is an integer.

Divisibility shortcuts & final-digit rule

Final digit	Final digit of a sum/product = final digit of the sum/product of the inputs' final digits.
Divisible by 2 / 5 / 10	Last digit even / last digit 5 or 0 / last digit 0.
Divisible by 3 / 9	Digit sum divisible by 3 / by 9.
Divisible by 4	Last two digits form a number divisible by 4.

3.2 Algebra, Equalities & Inequalities

Terms & expressions	A term is a constant, variable, or product of these; a coefficient is the constant multiplying a variable. Simplify by combining like terms and factoring out common factors. To multiply expressions, multiply every term of one by every term of the other.
Linear equation (1 unknown)	Isolate the variable by doing the same operation to both sides. Has one solution.
Two linear equations, two unknowns	Solve by substitution or by elimination (scale to match a coefficient, then add/subtract). Equivalent equations → infinitely many solutions (you'll hit 0 = 0); non-equivalent, non-parallel → exactly one solution; a contradiction → no solution.
Factoring identities	$a^2 – b^2 = (a+b)(a–b)$; $(a±b)^2 = a^2 ± 2ab + b^2$.
Solving by factoring	Move everything to one side (= 0), factor, set each factor to 0. A fraction = 0 only when its numerator = 0 and denominator ≠ 0.
Quadratic equation	Standard form $ax^2 + bx + c = 0$ ($a \neq 0$). At most two real roots.
Quadratic formula	$x = [–b ± \sqrt{(b^2 – 4ac)}] / 2a$. Discriminant $b^2–4ac$: > 0 → two roots, = 0 → one root, < 0 → no real root.
Inequalities	Solve like equations, BUT reverse the sign when multiplying/dividing by a negative. Never multiply by a variable whose sign is unknown.
Functions	f(x): substitute the input. One input → at most one output (different inputs may share an output). Domain = allowed inputs (no ÷0, no √ of negatives); range = set of outputs.
Quadratic function	$f(x) = ax^2+bx+c$. If a > 0 it has a minimum, if a < 0 a maximum, of value $c – b^2/4a$, occurring at $x = –b/2a$.

Coordinate geometry (lines only)

Line equation	y = mx + b: m is the slope, b is the y-intercept. Vertical line: x = a (undefined slope); horizontal: y = b (slope 0).
Slope	m = (y■ – y■)/(x■ – x■). Positive → rises L→R; negative → falls; 0 → horizontal. Keep the point order consistent in numerator and denominator.
Point-slope	Through (x■,y■) with slope m: y – y■ = m(x – x■).
Intercepts	x-intercepts solve f(x)=0; y-intercept is f(0). Two lines: unique solution → intersect; equivalent → same line; no solution → parallel.
Unit conversion	A GMAT question requiring a conversion (except time units) will give you the relationship. Translate it into a formula and apply.

3.3 Rates, Ratios & Percents

Ratio & proportion	Ratio a:b = a/b; order matters. A proportion sets two ratios equal; solve by cross-multiplying. a:b:c splits a total into (a+b+c) parts.
Fractions	Reduce by dividing out the GCD. Same denominator → add/subtract numerators. Different → find a common denominator first. Multiply: numerators and denominators across. Divide: multiply by the reciprocal.

Percent	"Per hundred." part = percent × whole. Percents can exceed 100% or be fractions of a percent. Convert: percent→decimal move point 2 left; decimal→percent move 2 right.
Percent change	(new – old)/old × 100. A 25% rise then needs only a 20% fall to return — increases and decreases are not symmetric. Successive % changes MULTIPLY (a 20% then 30% discount = 0.8×0.7 = 56% of original, a 44% total discount).
Discount / profit	Price after p% discount = (100–p)% of original. Profit = revenue – expenses = selling price – total cost.
Simple interest	I = principal × rate × time.
Compound interest	$A = P(1 + r/n)^{nt}$, where r = annual rate, n = compounds per year, t = years. Each period's interest is added to the balance before the next.
Distance	distance = rate × time.
Average speed	= total distance ÷ total time — NOT the average of the speeds.
Work	1/a + 1/b = 1/T, where a, b are solo times and T is the combined time. Rearrange to find any unknown.
Mixtures	Track the quantity of each component (e.g. amount of salt), not just the percentages; set the totals equal.

3.4 Statistics, Sets, Counting, Probability, Estimation & Series

Descriptive statistics

Mean	Sum of values ÷ count. Each repeated value is counted each time.
Median	Order the values; middle one if the count is odd, average of the two middle ones if even. May be above, below, or equal to the mean; resists outliers.
Mode	Most frequent value; a data set can have more than one mode (or none).
Range	Greatest value – least value; depends on only two values.
Standard deviation	Spread about the mean: (1) find the mean, (2) find each value's difference from it, (3) square the differences, (4) average them, (5) take the square root. Larger SD = more spread. Values farther from the mean affect it most; tightly clustered data has a small SD.
Frequency distribution	A table of each value and how often it occurs; speeds up mean/median/mode.

Sets & counting

Set basics	A collection of distinct elements; order doesn't matter. |A| = number of elements. A is a subset of B if every element of A is also in B.
Union / intersection	Union = in A or B or both; intersection = in both. Disjoint (mutually exclusive) sets share no elements; the empty set is { }.
Addition rule	n(A or B) = n(A) + n(B) – n(A and B). Solve overlapping-group word problems with a Venn diagram or this rule.
Multiplication principle	Independent stages with m, n, … choices give m × n × … total outcomes. Choosing 1 from each of k sets of r objects = r^k.
Factorial	n! = n × (n–1) × … × 1; 0! = 1! = 1. Note n! = n × (n–1)!.
Permutations	Orderings of n objects = n!. Order matters.
Combinations	Choosing k from n with order not mattering: nCk = n! / [k!(n–k)!]. Note nCk = nC(n–k).

Probability

Definition	P(event) is between 0 and 1. For equally likely outcomes, P = favorable ÷ total. P(impossible)=0, P(certain)=1.
Complement	P(not A) = 1 – P(A).
Addition rule	P(A or B) = P(A) + P(B) – P(A and B). If mutually exclusive, P(A and B)=0, so P(A or B) = P(A) + P(B).
Independent events	Neither changes the other's probability: P(A and B) = P(A)·P(B).
Dependent events	General rule P(A and B) = P(A)·P(B given A). A is dependent on B when P(A given B) ≠ P(A).

Estimation & sequences

Estimation	Round inputs to fewer digits to approximate fast; keep more digits for more accuracy. Sometimes round to the nearest square/cube. Bounding (finding an upper and lower bound) can beat a single estimate.
Arithmetic sequence	Each term adds a constant d: a■ = a■ + (n–1)d.
Geometric sequence	Each term multiplies by a constant r: a■ = a■·$r^{(n-1)}$.
Recursive / series	A sequence may be defined by its first term(s) plus a rule for the next. A series is the sum of a sequence's terms; some infinite series sum to a finite value, many don't.

Quant strategy & common traps (Ch. 4)

- First read what's given and what's asked; scan the choices. If your answer isn't among them, re-read the question.
- No calculator — estimate, round, and look for arithmetic shortcuts; the numbers usually work out cleanly.
- When algebra is messy, plug in smart numbers or test the answer choices.
- Answer the EXACT quantity asked (x vs. x^2, vs. an expression's value). Mind units and whether a count/fraction/percent is wanted.
- Use only the information given; don't apply outside subject knowledge. All needed formulas/conversions are provided.

2 Data Insights

Official Guide Ch. 5 (DI Review) & Ch. 6 — reason with data in real-world formats; on-screen calculator allowed

5.1 Data Sets & Types of Data

Key terms	A case is one individual/thing the data describes; a variable is a type of data held about each case; a value is that variable for one case; a data point is one value in one case (a cell); a record is all data points for one case (a row).
Independent vs dependent	A dependent variable's value is computed from others (e.g. profit = revenue – expense); independent variables aren't.

Qualitative (categorical) data — which statistics apply

Type	Meaning	Mean	Median	Mode	Range
Nominal	Unordered categories (names, phone numbers, colors)	no	no	yes	no
Ordinal	Ordered but not numeric (weekdays, rankings)	no	yes	yes	no
Binary	Only two values (yes/no, T/F); ordinal if order matters, else nominal	—	—	yes	—
Partly ordered	Order among some cases only (e.g. a family tree by age)	no	no*	yes	no

**median may apply to a fully-ordered subset. Data that looks numeric is still qualitative if the numbers don't represent quantities (e.g. phone numbers).*

Quantitative (numeric) data

Continuous vs discrete	Continuous = infinitely divisible (temperature, altitude) and rarely has a mode; discrete = countable/whole or limited to fixed steps (number of students, prices to the cent).
Interval	Zero does NOT mean “none” (Celsius/Fahrenheit, calendar years). Differences are meaningful, but ratios are not — 60°F is not “twice as hot” as 30°F.
Ratio	Zero means a true absence (weight, Kelvin). Both differences and ratios are meaningful — 200 K really is twice 100 K.
Logarithmic	Each step is an exponential jump (decibels). Can't just sum/divide to get a mean, or subtract for a range (not tested by hand).

5.2 Data Displays

Always read the title, axis labels, units, scales, and legend first; never assume a graphic is to scale.

Tables & qualitative charts

Tables	Header row names the variables; rows are records, columns are variables, cells are data points. Watch for grouping rows and total/average rows, and for units in the title (e.g. “in thousands”).
Network diagram	Nodes = individuals, lines = relationships; arrows show direction (one-way / two-way).
Tree diagram	A one-way network for partly ordered data (org charts, ancestry, conditional probabilities).
Flowchart	Steps in a process: ovals = start/end, rectangles = actions, diamonds = decisions with labeled branches; a back-arrow means repeat.
Gantt chart	Project schedule: horizontal = time, each row = a task with a bar from start to end; arrows show one task must finish before another starts.

Quantitative charts

Pie chart	Slices are proportions of a whole, summing to 100%; use the fractions/percents in calculations.
Bar chart	Bar height/length = a value per case. Grouped = bars clustered to compare variables/cases; stacked = segments within a bar are parts of a whole.
Histogram	Bars represent ranges of one variable's values (not separate cases); height = frequency in that range. Bars are ordered low→high and don't overlap.
Line chart	Usually values over time; great for trends and correlations. Two variables may use two different vertical scales (left and right).
Scatterplot	One quantitative variable per axis; each dot is a case. Shows correlation and how tightly points fit a trend line. Bubble charts add a third variable via dot size.

5.3 Data Patterns

Distributions	Uniform = all values about equally frequent. Normal = symmetric bell; mean = median = mode at the center. Skewed = one longer tail; the mean is pulled toward the long tail, past the median.
Spread	Tighter clustering → taller, narrower peak → smaller SD → values more predictable. Wider spread → larger SD. Distributions can also be bimodal or irregular.
Trends	A trend observed over a period is evidence it extends a little beyond — stronger the longer and more varied the observed period, weaker the farther out you extrapolate.
Correlation	Positive = both rise together (lines slope together; scatter clusters around an up-sloping line). Negative = one rises as the other falls. Tighter clustering / more consistent slopes = stronger correlation.
Nominal association	An unordered category (e.g. species, major) can still be associated with higher/lower values of another variable — an association, not a positive/negative correlation.
Correlation ≠ causation	A correlation can arise from coincidence or a hidden third factor. Test causation by intervening (stop the suspected cause and see if the effect stops) and by ruling out confounders.

6.1 What Is Measured

Skill	What you do
Apply	Use a rule/principle in a new context; say what follows if new information is added.
Evaluate	Judge whether evidence supports or weakens a claim or plan; find gaps or errors.
Infer	Draw unstated but logically supported conclusions; find probabilities, rates of change, meaning in context.
Recognize	Identify what's stated explicitly; spot agreements/conflicts between sources; gauge correlation strength.
Strategize	Find a course of action that meets constraints; weigh trade-offs; pick the formula/plan that hits a goal.

6.2 The Five Question Types & Strategies

Multi-Source Reasoning	2–3 tabs (text, tables, graphs) on the left; 3 questions appear one at a time. Some are 5-choice multiple choice; others are 3-row yes/no (conditional) where ALL three rows must be right for credit. Note each statement's role; click between tabs freely; some questions ask you to spot conflicts, others to combine sources.
Table Analysis	One sortable table plus 3-row yes/no statements (all rows must be correct). Read the surrounding text to learn what the data is; SORT by the column a row hints at to find the answer fast; match each row against the stated condition ("consistent with", "can be inferred").
Graphics Interpretation	A graphic plus sentences with drop-down blanks; fill every blank for credit. Study axes, scales and units (graphic units may differ from the text); read all menu choices; for "nearest to / closest to", pick the option closest to your calculation.

Two-Part Analysis	A passage and a 3-column response table: pick ONE answer in each of the first two columns (per column, not per row). The same option may be correct for both. Read the instructions — the two choices may be independent or must combine into one answer; read all third-column options first.
Data Sufficiency	Decide whether the statements give ENOUGH information — you do NOT solve. Same 5 answer choices every time (below).

Data Sufficiency — the five fixed answer choices (memorize cold)	
A	Statement (1) ALONE is sufficient, but (2) alone is not.
B	Statement (2) ALONE is sufficient, but (1) alone is not.
C	BOTH statements TOGETHER are sufficient, but NEITHER alone is.
D	EACH statement ALONE is sufficient.
E	Statements (1) and (2) TOGETHER are still NOT sufficient.

Data Sufficiency decision routine: Evaluate statement (1) ALONE — sufficient? Then evaluate statement (2) ALONE, ignoring (1) entirely. If exactly one works → A or B. If each works on its own → D. If neither alone works, test them TOGETHER → C if that's now enough, E if not. For a value question, “sufficient” means the data pin down exactly one number.

Data Insights strategy & common traps (Ch. 6)

- Data Sufficiency: never carry information from statement (1) into your check of (2); judge each alone first.
- Don't waste time actually solving a DS problem — only decide whether a unique answer is possible.
- The 'C trap': combining always feels safe, but check whether one statement alone already suffices (answer D).
- On multi-question screens you can't go back once you click Next — finalize all answers on that screen first.
- Correlation isn't causation; a single outlier can distort an average or trend. Use the on-screen calculator for arithmetic.

3 Verbal Reasoning

Official Guide Ch. 7 (Verbal Review) & Ch. 8 — Reading Comprehension & Critical Reasoning

7.1 Analyzing Passages

Argument	Gives idea(s) as reasons to accept other idea(s); some ideas may be implied. Built from premises and conclusions.
Premise	An idea offered as a reason. Cue words: because, since, for, given that, after all, for the reason that, in light of, furthermore, moreover.
Conclusion	An idea the premises support. A main conclusion supports no further conclusion; an intermediate conclusion supports another. Cue words: therefore, thus, hence, so, clearly, surely, it follows that, shows/implies/suggests/proves that. (When there are no cue words, ask which statements are given as reasons for which.)
Valid vs sound	Valid = the conclusion follows IF the premises are true (can still have false premises). Sound = valid AND premises actually true → conclusion must be true.
Assumption	An idea taken for granted; never the conclusion. A necessary assumption MUST be true for the argument to work (test by negating it — the argument should collapse). A sufficient assumption, if true, makes the conclusion follow.

Types of conclusion / argument

Prescriptive	Says what should/shouldn't be done (policies, actions).
Evaluative	Says something is good/bad, desirable/undesirable, without advocating an action.
Interpretive	About the meaning, significance, or implications of something.
Causal	Claims a factor did/didn't contribute to an effect.
Factual	A plain factual conclusion fitting none of the above.

Non-arguments (know the difference)

Causal explanation	Says what causes an effect, without arguing the effect is real. Cues: because, causes, leads to, results in, due to, that's why, thereby. Can be a premise/conclusion inside an argument, but isn't itself one.
Observation / hypothesis	An observation is directly known; a hypothesis is tentative (neither known nor assumed). Passages may weigh competing hypotheses without endorsing one.
Plan	A set of actions to reach a goal — not an argument (actions aren't reasons to accept the goal). Plans have necessary/sufficient assumptions just like arguments.
Narrative / description	A sequence of events, or a report of views/findings/places, with no conclusion argued. Statements can also give background, detail, the author's attitude, examples, or summarize an opposing view.

7.2 Inductive Reasoning

Premises support a conclusion as **probable**, not certain. Strengthen by adding support for the conclusion or its link to the premises; weaken by casting doubt on either.

Generalization	From a sample to a population (or population to part, or observed to unobserved). Strength rises with sample size and with how representative the sample is.
Biased sample	A sample likely to differ from the population (e.g. a phone survey of who answers phones). Any argument built on it is flawed.

Hasty generalization	A sample too small to justify the conclusion (5 of 8 coin flips ≠ a weighted coin).
Fallacy of specificity	A conclusion too precise for the evidence (50 frogs averaging 32.86 g ≠ all frogs averaging exactly 32.86 g). A vaguer conclusion is better supported by the same evidence.
Causal reasoning	A correlation supports a cause only after ruling out (a) reverse causation, (b) a hidden common cause, and (c) coincidence. Test by isolating one suspected cause at a time, without adding new factors.
Analogy	Argues two things alike in some ways are alike in another. Strong only if the noted similarities are RELEVANT to the conclusion; strengthen with more relevant similarities, weaken with relevant differences.

7.3 Deductive Reasoning

Premises are offered to fully **prove** the conclusion. A valid deductive argument with true premises must have a true conclusion; it's flawed if the premises can be true while the conclusion is false.

Negation	“not A” is true exactly when A is false.
Conjunction (and)	“A and B” true only when both are. Markers: and, but, although, however, whereas, even though (the latter signal tension/surprise).
Disjunction (or)	Inclusive “A or B or both” = at least one true; exclusive “A or B but not both” = exactly one. “A unless B” marks a disjunction.
Conditional (if-then)	“If A, then B” means A requires B; A isn't claimed true and isn't a premise. Equivalent to its contrapositive “if not-B, then not-A.” It does NOT imply “if B, then A.”
Biconditional	“A if and only if B” — each implies the other; both always true or both false.

Valid vs. invalid forms (memorize)

Valid	Invalid (common trap)
If A then B; A is true → B (modus ponens)	If A then B; B is true → A (affirming the consequent)
If A then B; not-B → not-A (modus tollens / contrapositive)	If A then B; not-A → not-B (denying the antecedent)
not-A and not-B → not (A and B)	not (A and B) → not-A and not-B
A and B → A; A → A or B	A → A and B; A or B → A

Modality & quantifiers

Necessity / probability / possibility	Necessarily (must, certainly) = 100%; probably (likely) = a good chance; possibly (may, could, might) = above 0%. Valid: “probably A → possibly A.” Invalid: “possibly A → probably A.” Don't assign exact percentages to probably/likely.
Quantifiers	all = 100%; most = more than half (usually implies not all); some = at least one/part (usually implies not all; “only some” does, “at least some” doesn't); none = 0. Watch the scope of statements built from quantifiers.

8.2 Reading Comprehension — the five question types

Main Idea	The passage's overall point, purpose, or theme. For an argument, it restates the main conclusion. Cues: “primary purpose”, “main idea”, “author seeks primarily to”. Wrong answers grab a detail, twist the point, or add outside ideas.
Supporting Idea	About any stated detail except the main point — a premise, objection, example, definition in context, etc. Answers paraphrase rather than quote. Cues: “according to the passage”, “the author cites/mentions/offers as an objection”.

Inference	An idea the passage supports but doesn't state. Must follow from the text (may follow from even a false statement in the passage). Cues: "it can be inferred", "the passage implies/suggests", "most strongly supported". Wrong answers are true/related but unsupported.
Application	Connects the passage to a NEW situation it doesn't mention — by analogy, by applying a rule, by extending the passage, or via a "what-if". Don't reject an answer just because it's about an unmentioned topic; that's expected here. Cues: would/could/might/should, "most similar to", "best exemplifies".
Evaluation	Assess the passage's structure and logic — a part's function, how the author persuades, an implicit assumption, what would strengthen/weaken/resolve a conflict. Answer choices are often abstract. Overlaps with Critical Reasoning.

8.2 Critical Reasoning — the four subtypes

Analysis	About the argument's structure and the role statements play (often boldface). Also covers argumentative method, or an unstated point/point of disagreement in a dialogue. Cues: "the two portions in boldface play which roles", "the argument proceeds by", "main point of disagreement".
Construction	Complete a partial argument/explanation: draw the conclusion the premises support, supply a missing premise (necessary or sufficient assumption, observation, or justification), or best explain a discrepancy. Cues: "most logically completes", "depends on the assumption that", "best explains the discrepancy", "enables the conclusion to be properly drawn".
Critique	Judge reasoning: name a flaw, find what most strengthens or weakens it, or say what it would be most useful to know. Strengthen/weaken items are always inductive. "Useful to know" answers start with "Whether…". Cues: "most vulnerable to the criticism that", "logically flawed in that", "casts the most serious doubt on".
Plan	Construct or judge reasoning about a course of action: what must be true for it to succeed, what would make it more/less likely, how it's flawed, what would best evaluate it. Same phrasings as Construction/Critique but always about a plan/strategy.

Core CR routine: find the conclusion → find the premises → identify the gap/assumption between them. Strengthen confirms the assumption or rules out an alternative cause; weaken attacks the assumption or offers an alternative explanation; an assumption is required (negate it to test); inference/"must be true" stays strictly within what's stated.

Verbal strategy & common traps (Ch. 8)

- Answer only from the passage plus common knowledge — not specialist outside knowledge of the topic.
- Watch cue words that mark a statement's role, and read the question stem precisely (subtle wording flips the answer).
- Extreme/absolute words (all, never, only, must) are often wrong in RC — prefer the supported, moderate choice.
- Beware answers true in the real world but not stated or implied by the passage.
- Read every choice; eliminate scope-shifts, reversals and irrelevant comparisons; if unsure, rule out and pick the best of what's left.

4 Night-Before Quick Checklist

A fast confidence pass — if any line feels shaky, flip back to that section

Quantitative

- Divisibility rules (2,3,4,5,9,10); primes to ~50; prime factorization → count of factors.
- Exponent & root rules; $\sqrt{x^2}=|x|$; negatives have no real square root.
- $a^2-b^2=(a+b)(a-b)$; $(a\pm b)^2$; quadratic formula and the discriminant test.
- Flip the inequality sign on multiply/divide by a negative.
- % change = (new–old)/old; successive percents multiply; average speed = total dist ÷ total time.
- Simple I=Prt; compound A=P(1+r/n)^nt; work 1/a+1/b=1/T.
- Mean/median/mode/range/SD; n(A or B)=n(A)+n(B)–n(A and B); nPr, nCk; P(not A)=1–P(A), independent multiply.

Data Insights

- Which stats apply to nominal/ordinal/quantitative data; interval vs ratio (zero meaning).
- Read titles, axes, units, scale, legend before answering; never assume a graphic is to scale.
- Normal vs skewed (mean follows the long tail); correlation ≠ causation.
- The five DI question types and their response formats; the five DS answer choices.
- DS routine: each statement ALONE first, then together; don't actually solve; beware the C-trap.

Verbal

- Premise vs conclusion cue words; valid vs sound; necessary vs sufficient assumption (negation test).
- Inductive flaws: biased sample, hasty generalization, fallacy of specificity, correlation≠causation, weak analogy.
- Conditional = contrapositive; traps: affirming the consequent, denying the antecedent.
- Quantifiers (all/most/some/none) and modality (necessarily/probably/possibly).
- RC types (Main Idea, Supporting, Inference, Application, Evaluation); CR types (Analysis, Construction, Critique, Plan).
- CR routine: conclusion → premises → gap; reject extreme and out-of-scope answers.

Exam-day reminders

- 64 questions, 135 minutes, three 45-minute sections; one optional 10-minute break.
- You can't skip; you may bookmark and edit up to 3 answers per section. Choose your section order.
- Pace: don't sink more than ~2 minutes into one question — eliminate, choose, move on.
- Total 205–805; sections 60–90. Rest well; trust your preparation.

Source: GMAT Official Guide 2025–2026, Graduate Management Admission Council (GMAC). Concept coverage drawn from Chapters 3, 5 and 7 (reviews) and the question-type strategies in Chapters 4, 6 and 8. Chapter numbers beside each heading point to where each topic is taught, with worked examples. This handbook is a revision aid — pair it with the Official Guide's practice questions and the Online Question Bank.